SPECIFICATIONS PLATFORM

Phase 1

UX Analysis, Information Architecture, User Flow & Sitemap

Deliverables 1–4 of 12

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Project **Malls Specs — reusable specification portal**

Reference system **Primedia Instore Media Specs (GitHub Pages)**

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Status **Awaiting approval to proceed to Phase 2**

> **SCOPE GUARD**
> This document analyses the reference system's framework only. No specification content, copy, dimensions, material names or product data from the reference site has been carried into this analysis or will be carried into the build. Every field in the new platform ships empty.

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1. Method & Evidence Base

The reference application was analysed at source rather than by inspection alone. The published document was retrieved in full, and its stylesheet, markup and behavioural layer were read directly. Every claim in this document traces to an observed declaration in that source — design tokens, breakpoints, component classes, transitions and state names are quoted as found, not inferred from screenshots.

1.1 What the reference system is

A single-document application. One HTML file carries the entire product: the design system as an inline stylesheet, the full page markup for every view, the behavioural layer, and the specification dataset itself as an inline JavaScript object. It is served statically from GitHub Pages. There is no build step, no framework and no backend.

One third-party dependency is loaded, deferred, from a CDN: jsPDF 2.5.1, used to generate the PDF export of a specification.

1.2 Consequences of that architecture

This shape explains most of what follows — both its strengths and its ceiling.

- **Strength:** zero infrastructure. It deploys by committing a file. Load is a single round trip. Nothing can break in a dependency chain because there is no dependency chain.
- **Strength:** the design system is disciplined. Twenty-two CSS custom properties govern colour, layout rhythm and motion; almost nothing is hard-coded outside them. That token layer transfers cleanly to a component architecture.
- **Ceiling:** content and application are the same artefact. Every specification is parsed and held in memory on first paint whether or not it is viewed. The document already exceeds 80 KB before the dataset is complete. At several hundred specifications this becomes the dominant cost of every page load.
- **Ceiling:** no view can be code-split, no image can be deferred, no search index can be built ahead of time, and no route can be pre-rendered — because there are no modules and no routes in the compiled sense.
- **Ceiling:** behaviour is bound through inline `onclick` attributes on markup. Interaction cannot be tested, reused, or given consistent keyboard and focus semantics.

> **CARRY FORWARD**
>
> The token layer and the interaction vocabulary are worth preserving almost exactly — they are the product's character.
>
> The delivery architecture is the thing being replaced. React, TypeScript and Vite exist in this brief to break the content-and-application coupling, not to modernise for its own sake.

2. UX Analysis

2.1 Overall information architecture

The reference system is three levels deep and organises entirely around a single content type.

```
Category          e.g. a top-level media family
  └ Sub-group     an optional intermediate grouping
    └ Spec        the leaf record — the only page with real content
```

Two structural observations matter for the rebuild.

- **The sub-group level is optional.** Some categories expose specifications directly; others insert a grouping tier. The navigation component handles both cases with the same primitive, distinguished by a `has-sub` modifier class. The data model must therefore treat the middle tier as nullable, not required.
- **Browsing lives entirely in the sidebar.** The landing view is a hero — a headline, a one-line description and three summary statistics. It contains no list, no filter panel and no results surface. To reach any specification the user must open the sidebar tree. This is the single largest divergence between the reference system and the platform described in your brief, and Section 5 addresses it directly.

2.2 Layout structure

A fixed application shell, not a scrolling document. The root element sets overflow to hidden on both html and body; only inner panes scroll. The shell is three fixed regions:

Region	Dimension	Behaviour
Top bar	52px fixed height	Fixed to viewport. Translucent white at 97% with a 24px backdrop blur, a hairline bottom border and a 1px shadow. Sits above everything at z-index 200.
Sidebar	260px fixed width	Solid navy. Independently scrollable with its scrollbar suppressed entirely. Does not shrink.
Main	Fills remainder	Overflow hidden, position relative. Acts as a stage on which views are absolutely positioned and cross-faded.

The brand block in the top bar is 260px wide and navy-filled, so it reads as the head of the sidebar column rather than as part of the bar. The two navy surfaces meet to form a single L-shaped chrome.

View switching

Home and the specification view both occupy the main stage absolutely, at inset zero. They are never mounted and unmounted — they are cross-faded by toggling classes.

- **Leaving home:** opacity to 0, scale to 0.98, translateY −8px, pointer-events off, over 350ms.
- **Entering a spec:** opacity 0 to 1 from translateY +10px, over 300ms.

The two overlap. The effect is that the spec view rises as home recedes, rather than one replacing the other. This is a signature interaction and should survive the rebuild — in React it becomes a route transition, not a class toggle.

The specification view itself

A two-pane split. The data panel takes exactly 50% of the stage; the image pane takes the remainder and is filled with a navy-backed, cover-fitted photograph. The panel is a vertical flex column with three parts: a fixed action bar at the top, a scrolling content region in the middle, and a pinned downloads bar at the bottom. Only the middle region scrolls.

2.3 Navigation behaviour

The sidebar is a filtered, single-select, accordion tree with four constituent parts.

Element	Role	Observed treatment
Search field	Filters the tree	32px tall, inset magnifier icon, 10% white fill on navy, 8px radius. Focus lifts the fill to 15% and the border to 28% white.
Section label	Groups categories	8.5px, 600 weight, uppercase, 0.14em tracking, 45% white. A flexed hairline rule fills the remaining width after the text — a pseudo-element, not a separate node.
Category item	Expands a branch	Full-width button, 7px radius, 12.5px text at 75% white. Where it has children it carries a chevron that rotates 90 degrees on open.
Sub-item	Selects a spec	Indented 16px, 12px text at 60% white, prefixed by a 3px dot.

Selection state

The selected item receives a 25%-opacity accent fill and a 2px accent bar pinned to its left edge at 60% of its height, drawn as a pseudo-element with an asymmetric radius. It is a precise, restrained treatment and is the clearest visual anchor in the entire interface. It must be reproduced exactly — and, critically, it must be paired with a real focus ring, which the reference system does not provide.

Expand and collapse

Branches animate open by transitioning max-height from 0 to a fixed 600px ceiling over 280ms. This is the standard CSS workaround for animating to an unknown height, and it carries two defects that the rebuild must not inherit.

- Any branch whose natural height exceeds 600px is silently clipped. With hundreds of specifications this is a matter of when, not if.
- Because the branch is hidden by overflow rather than removed, its buttons remain in the tab order while collapsed. A keyboard user tabs into items they cannot see.

2.4 Search behaviour

Search is a client-side filter over the navigation tree, scoped to the sidebar. It has no results page, no result count, no empty state, no keyboard shortcut and no ability to reach anything other than a specification title. It does not search descriptions, specification rows, reference codes or categories.

For a portal of the size described in your brief this is the weakest surface in the system. Section 5 proposes replacing it with a two-tier model: the sidebar filter is retained for orientation, and a global command-palette search is added over a pre-built index.

2.5 Filtering behaviour

There is no filtering in the public interface. The category tree is the only mechanism for narrowing, and it is a navigation control, not a filter — it selects one branch rather than combining criteria. Filtering exists only inside the admin panel, where a 200px search input performs a substring match over the specification list.

Your brief specifies a filter panel, category filters, tags and status. None of these have a precedent in the reference system; they are net-new architecture and are designed from first principles in Phase 3.

2.6 Component inventory

Extracted from the source. Marked components are those that exist today; the gap column notes what your brief requires beyond them.

Group	Present in reference	Required additions
Chrome	Top bar, brand block, breadcrumb strip, sidebar, mobile nav toggle, backdrop scrim	Skip link, footer, command palette trigger
Navigation	Search input, section label, category item, chevron, sub-item, selection marker	Anchor rail, pagination, recently-viewed list, bookmarks list
Buttons	Top-bar button, back button, download bar button (solid and ghost), admin button, admin action button, destructive variant, icon-only variant	Icon button with accessible name, toggle button, link button, loading state
Data display	Reference-code chip, statistic tile, spec block with rule, bulleted spec row, sub-heading row, two-column boxed group, file row	Status badge, tag chip, definition table, revision-history table, related-spec card, attachment card
Overlays	Curtain scrim, sheet, admin sheet, auth modal, spec form sheet, notification panel	Tooltip, popover, confirmation dialog, toast
Forms	Text input, select, textarea, inline category creator, image drop zone, PDF drop zone, toggle	Combobox, tag input, checkbox, radio group, field error, help text
Feedback	Notification list, unread count badge, image placeholder	Empty state, loading state, error state, skeleton loader, progress indicator
Admin	Tab bar, toolbar, grid list with header row, category accordion, group row, count label	Bulk actions, sort control, pagination

2.7 Visual hierarchy

Hierarchy is carried by three devices used consistently.

1. Extreme type contrast. The home headline runs to a clamped maximum of 100px at weight 900 with −0.04em tracking and 0.92 line-height, against a 13px body. That is a ratio near 8:1 — far beyond conventional documentation practice, and it is what gives the landing view its authority.
2. Colour as rank, not decoration. Navy is reserved for headings and structural chrome. The accent blue appears only on interactive or referential elements — selected navigation, reference codes, the code chip, spec row bullets. Body text is near-black; supporting text steps down through two alpha levels. Nothing is coloured for its own sake.
3. Label-plus-rule section headers. Every block in the spec view is introduced by an 11px uppercase label at 0.14em tracking with a hairline rule flexing to fill the line. It costs one pseudo-element and does the work of a much heavier heading.

2.8 Typography system

Two families. Wix Madefor Display carries every heading, label, numeral and chip. Montserrat carries body copy, controls and navigation. The split is consistently observed — display type never appears in running text and body type never appears in a heading.

Role	Family	Size	Weight	Tracking / leading
Home headline	Display	clamp(52–100px)	900	−0.04em / 0.92
Statistic numeral	Display	24px	900	−0.03em / 1.0
Spec title	Display	26px	800	−0.02em / 1.15
Section label	Display	11px	700	0.14em, uppercase
Nav section label	Display	8.5px	600	0.14em, uppercase
Code chip	Display	9px	600	0.06em

Role	Family	Size	Weight	Tracking / leading
Breadcrumb	Display	10px	400	0.06em
Table header	Display	8px	700	0.14em, uppercase
Body / base	Body	13px	400	—
Spec description	Body	13px	300	— / 1.75
Spec row	Body	12.5px	400	— / 1.6
Nav item	Body	12.5px	500	— / 1.0
Sub-item / control	Body	12px	400–500	—
Statistic label	Body	10px	400	0.1em, uppercase

ACCESSIBILITY FINDING — MUST BE CORRECTED

The scale bottoms out at 8px. Six distinct roles sit at or below 10px, including the primary navigation section labels and every admin table header. Below roughly 11px, uppercase text at wide tracking becomes materially harder to read and does not survive user zoom well.

The new scale keeps the same visual rhythm and the same contrast ratio between roles, but rebases the floor at 11px and expresses every step in rem so it responds to user font-size preferences. Nothing about the character of the interface is lost; it simply stops excluding people.

2.9 Colour system

Twenty-two tokens, built on a single hue relationship: a navy anchor, a brighter blue accent, and a neutral ramp derived from the navy rather than from grey. Every surface, border and text tint is an alpha of the navy over white, which is why the interface reads as tonally unified rather than assembled.

Token	Value	Applied to
--bg	#FFFFFF	Page and panel base
--bg1	#F4F6FB	Boxed sub-groups, hover fills
--bg2	#EEF1F8	File rows, table header rows
--bg3	#E4E9F4	Control hover, admin search fill
--line	navy @ 8%	Hairlines, section rules
--line2	navy @ 14%	Control borders, dividers
--line3	navy @ 22%	Hover borders, scrollbar thumb
--txt	#0D1E3F	Primary text
--txt2	ink @ 55%	Secondary text, control labels
--txt3	ink @ 35%	Tertiary text, placeholders
--acc	#2861FD	Interactive, selected, referential
--acc2	#1A4FD6	Accent pressed / hover
--acc-dim	accent @ 10%	Chip and active-tab fills
--navy	#1B3769	Sidebar, brand, headings, image pane

Token	Value	Applied to
--red	#FF4D4D	Destructive actions only
--blue	#4D8FFF	Secondary accent

ACCESSIBILITY FINDING — MUST BE CORRECTED

--txt3 (ink at 35%) on white resolves to roughly 2.6:1. WCAG 2.1 AA requires 4.5:1 for body text and 3:1 for large text. This token is used for placeholders, statistic labels, breadcrumb text and admin metadata — it fails in every one of those positions.

--txt2 at 55% sits near 4.4:1: marginal, and it fails at the small sizes it is currently paired with.

The corrected ramp holds the same perceived tonal steps by darkening the underlying ink rather than by raising opacity, so the interface looks unchanged while passing. Dark mode is derived from the same token names, which is why the rebuild is dark-mode-ready by construction rather than by retrofit.

2.10 Spacing, grid and radii

There is no declared spacing scale, but the values used are highly regular — they resolve cleanly onto a 4px base with a small set of half-steps. This is fortunate: it means a formal scale can be imposed in Phase 3 with almost no visual drift.

- **Spacing values observed:** 2, 4, 5, 6, 7, 8, 10, 12, 14, 16, 20, 22, 24, 26, 28, 32, 44. All but 5, 7, 14, 22 and 26 are multiples of 4.
- **Control heights:** 22 (chip), 26 (admin button), 28 (back button, tab), 30 (top-bar button), 32 (all inputs, download button), 35 (form field).
- **Radii:** 5, 6, 7, 8, 10, 14 and 100px for pills. Radius scales with the element — small controls take 5–7, containers take 8–10, sheets take 14.
- **Grid:** no page grid. Layout is flexbox throughout, with CSS Grid used in exactly three places — the two-column spec boxes, the admin list rows and the spec form. The decorative grid on the home view is a 40px background pattern, not a layout system.

2.11 Interaction patterns and motion

Two easing curves do all the work, and the distinction between them is deliberate.

Curve	Definition	Used for	Character
--ease	cubic-bezier(.16, 1, .3, 1)	View transitions, accordions, image zoom, panel slides	Fast out, long settle. Expressive.
--ease2	cubic-bezier(.4, 0, .2, 1)	Hover, focus, button and nav state changes	Symmetrical, brief. Utilitarian.

Durations cluster tightly: 120–180ms for state changes, 280–350ms for structural movement, and a single 700ms outlier on the specification image zoom. Motion is used sparingly and never gratuitously — the only looping animation in the product is a two-second opacity pulse on a 5px dot in the home kicker.

Hover treatments are consistent: fill deepens one step on the neutral ramp, border deepens one step, text lifts one level toward primary. The download button additionally lifts 1px on the Y axis. Nothing scales, nothing bounces.

ACCESSIBILITY FINDING — MUST BE CORRECTED

No prefers-reduced-motion query exists anywhere in the stylesheet. Every transition described above plays unconditionally.

> No focus-visible styling is declared. Several controls suppress the default outline through border and background resets without replacing it, so keyboard focus is invisible in parts of the interface.
>
> Both are addressed as design-system primitives in Phase 3 rather than as per-component patches.

2.12 Status indicators, badges and icons

The reference system has no status concept. Nothing is draft, in review, published, deprecated or superseded — a specification simply exists, or it has been archived, and archive is a hidden administrative state rather than a visible status. Your brief requires status as a first-class, filterable, visible property; it is designed new.

The nearest existing precedent is the reference-code chip: a 22px pill in accent-dimmed fill with an accent border and 9px display type. It is the correct visual foundation for a badge family, and the status, tag and category badges in Phase 3 are derived from it so that the new elements look native rather than imported.

Icons are inline SVG at 11–14px, stroked, and set at 70% opacity so they recede against their label. There is no icon component and no sprite — each icon is written out at its point of use. The rebuild introduces a single typed icon component with a required accessible name.

2.13 Responsive behaviour

Four breakpoints, each with a clearly defined job.

Breakpoint	What changes
≤ 1024px	Sidebar narrows from 260px to 220px. The specification view rotates from a horizontal split to a vertical stack, with the image promoted above the data panel at a fixed 220px height and its focal point shifted to 30% from the top. The pane becomes document-scrolling rather than pane-scrolling. Home headline scale reduces.
≤ 768px	Top bar shortens to 48px. The sidebar leaves the flow entirely: it becomes a fixed 260px overlay translated off-canvas, restored by a hamburger toggle, with a scrim behind it and a 4px shadow along its edge. The spec action bar begins wrapping to two rows. Home switches from vertically centred to bottom-aligned. All modals go full-bleed and the spec form drops to a single column.
≤ 480px	Top bar shortens to 44px. The brand block gives up its fixed width. Button labels are hidden, leaving icon-only controls. The breadcrumb's specification segment is removed. The image pane shortens to 180px. Two-column spec boxes collapse to one. Statistic tiles distribute evenly and centre their content. File rows stack vertically.
≤ 360px	Top bar shortens to 42px. Home headline drops to a fixed 28px. The image download control is removed entirely.

The strategy is sound and the transitions are well judged. Three problems accompany it.

- The off-canvas sidebar is translated, not removed, so its controls stay focusable while closed — the same defect as the accordion, at a larger scale.
- Progressive removal of controls at 480px and 360px means small-screen users lose functionality outright rather than reaching it another way. The rebuild moves removed actions into an overflow menu instead.
- Overflow hidden on html and body prevents the reflow that WCAG 1.4.10 requires at 400% zoom.

2.14 Cards, tables and lists

Worth stating plainly: there are no cards in the reference system. There is no grid of specification tiles, no summary card, no related-item card. The only container that resembles one is the boxed sub-group inside the spec view — an 8px-radius panel in the lightest tint with a hairline border and a tiny uppercase label. Your brief's specification list,

related specifications and attachment surfaces all need a card component that has no precedent here; it is designed in Phase 3 from the boxed sub-group's materials.

Tables appear only in the admin panel, and they are not table elements — they are CSS Grid rows with a fixed four-column template: two flexible columns, one flexible column and a fixed 170px action column. Cells truncate with ellipsis and do not wrap. The header row is distinguished by fill and by 8px uppercase type. There is no sorting, no pagination and no bulk selection. Semantically this should be a real table so that screen readers can announce row and column relationships.

Lists are the dominant content structure. The specification body is built almost entirely from bulleted rows: a 4px accent dot at 50% opacity, baseline-aligned against 12.5px text at 1.6 leading, with an optional inline bold-accent reference span and an optional bold sub-heading row breaking the list into labelled runs. This is the most content-bearing pattern in the product and it transfers to the new system unchanged.

2.15 The administrative layer

Substantial, and often overlooked in a surface reading of the site. It comprises:

- An authentication modal with a sign-out control and an admin-mode toggle that carries its own active styling.
- A five-tab management sheet — Specs, Categories, Users, Archive, Settings — at a maximum of 960px, going full-screen below 768px.
- A specification editor with a name field, category select with inline category creation, sub-group select, description, three multi-line specification fields that parse one item per line and treat a leading double-asterisk as a sub-heading, two independent two-column toggles, a packaging field, an image drop zone and a multi-file PDF drop zone.
- A category manager with nested group rows and per-row actions.
- An archive tab, implying soft deletion rather than destructive deletion.
- A settings tab holding a GitHub personal access token, stored in the browser, used to commit die-line PDFs back to the repository. Save, clear and test actions are provided.
- A notification system: a bell with an unread count, a panel with a mark-all-read action, and an entrance animation.

Per your Phase 1 decision, all of this is rebuilt. The GitHub token mechanism is the one element that should not be reproduced as-is — a token with repository write scope held in browser storage is a meaningful security exposure. Phase 5 proposes an abstracted upload service with the token strategy as one swappable adapter behind it.

3. Findings Summary

3.1 Keep — the reference system's real assets

- The navy-and-accent token architecture, and the discipline of deriving every neutral from the anchor hue.
- The two-curve motion vocabulary and its expressive-versus-utilitarian split.
- The label-plus-flexing-rule section header.
- The bulleted specification row with inline reference highlighting and sub-heading breaks.
- The navigation selection marker — accent fill plus a left edge bar.
- The split-pane specification view and its stacking behaviour, which you have confirmed as the direction.
- The pinned downloads bar, which keeps the primary action permanently reachable.
- The reference-code chip as the basis for a badge family.
- The L-shaped navy chrome formed by the brand block meeting the sidebar.

3.2 Fix — defects to correct in the rebuild

Finding	Severity	Resolution
Tertiary text fails WCAG AA contrast at roughly 2.6:1	High	Rebase the neutral ramp on darker ink; verify every pairing programmatically
Type scale bottoms out at 8px across six roles	High	Rebase floor at 11px, express in rem, preserve relative rhythm
No focus-visible styling; default outlines suppressed	High	Focus ring as a design-system primitive on every interactive element
Collapsed accordions and closed off-canvas nav stay focusable	High	Conditional rendering plus inert; never hide by overflow alone
overflow:hidden on html and body blocks zoom reflow	High	Shell scrolls internally without locking the document
No prefers-reduced-motion handling	Medium	Global reduced-motion query plus a motion-safe token layer
Accordion max-height ceiling clips branches over 600px	Medium	Measured-height or grid-rows animation with no ceiling
Admin grid rows are not semantic tables	Medium	Real table semantics with header association and sortable columns
Controls removed rather than relocated on small screens	Medium	Overflow menu preserves every action at every width
Behaviour bound through inline onclick attributes	Medium	Typed event handlers in components; no inline handlers
Repository-scoped token held in browser storage	Medium	Abstract behind an upload service; token adapter is one option
Icons written out at each point of use	Low	Single typed icon component with a required accessible name
Images load eagerly with no dimensions reserved	Low	Lazy loading, explicit aspect ratio, blurred placeholder

3.3 Add — required by the brief, absent from the reference

These have no precedent to reproduce. They are designed from first principles and are the substance of Phases 2 and 3.

- **Browsable landing surface —** search, filter panel, specification list, footer. The reference landing is a hero with no browse affordance at all.
- **Status —** a visible, filterable lifecycle property on every specification.
- **Tags —** free cross-cutting classification, independent of the category tree.
- **Revision history —** versioned change records per specification.
- **Structured metadata, requirements and notes —** as distinct, typed sections rather than free-form specification blocks.
- **Attachments and downloads —** generalised beyond the reference system's PDF-only die-line handling, to images, video, PDF and CAD.
- **Related specifications —** explicit cross-references between records.
- **Anchor navigation —** an on-this-page rail with scroll-spy for long specifications.
- **Deep breadcrumbs —** the existing strip is decorative and truncates on mobile; it becomes a real, navigable trail.
- **Print view —** a dedicated print stylesheet, distinct from the jsPDF export.
- **Recently viewed, bookmarks and favourites —** persisted personal surfaces.
- **Global keyboard navigation —** a command palette, skip links, roving tab index in the tree, and modal focus traps.
- **Dark mode —** a second token set under the same names.
- **Empty, loading, error and skeleton states —** particularly important here, because every specification ships empty. These states are not an edge case in this platform; for a period they are the primary experience.

4. Information Architecture

The target architecture keeps the reference system's three-tier spine — it is correct, and it is what your content is shaped like — and adds two cross-cutting classification axes that do not participate in the hierarchy.

4.1 Content model hierarchy

```
Specification Portal
│
├── Category                primary axis · exclusive · ordered
│   ├── Sub-group           optional middle tier · nullable
│   │   └── Specification   leaf record
│   └── Specification       categories may hold specs directly
│
├── Tag                     cross-cutting · many-to-many · unordered
├── Status                  lifecycle · exclusive · ordered
└── Collections             derived, never authored
    ├── Recently viewed       session/device scoped
    ├── Bookmarks             device scoped
    └── Favourites            device scoped
```

The distinction between the three axes is the load-bearing idea. Category answers "where does this live"; it is exclusive, ordered and navigable. Tag answers "what is this like"; it is many-to-many, unordered and filterable but never navigable as a tree. Status answers "can I rely on this"; it is exclusive, ordered and both filterable and displayed. Collapsing any two of these into one — the usual failure mode in specification portals — is what makes them unmaintainable at scale.

4.2 Specification page section order

Sequenced by decreasing certainty of need. A user arriving at a specification almost always wants identity and technical detail; they less often want history. The order below reflects that, and the anchor rail makes the lower sections cheap to reach.

#	Section	Placement	Empty-state treatment
1	Identity — title, code, status, category trail	Action bar and content head	Title placeholder; status defaults to Draft
2	Description	Content head	Placeholder line
3	Tags	Below description	Hidden when empty
4	Media	Right pane; above panel when stacked	Placeholder frame with upload affordance
5	Technical specifications	Content, first block	Em dash under an empty block label
6	Requirements	Content	Em dash under an empty block label
7	Metadata	Content, boxed	Definition rows with em-dash values
8	Notes	Content	Em dash under an empty block label
9	Attachments	Content	Empty state with add affordance
10	Downloads	Pinned bottom bar	"No files available"
11	Related specifications	Content, near end	Hidden when empty

#	Section	Placement	Empty-state treatment
12	Revision history	Content, last	Empty state — "No revisions recorded"

4.3 Navigation model

Four coordinated systems, each with a distinct job. Keeping them distinct is what prevents the navigation from becoming ambiguous as the catalogue grows.

System	Location	Answers	Scope
Category tree	Sidebar, persistent	Where am I in the catalogue?	All specifications
Global search	Top bar and ⌘K palette	Where is the thing I already know exists?	Titles, codes, descriptions, tags, body
Filter panel	Landing and category views	Which specifications match my criteria?	Current result set
Anchor rail	Specification view	Where am I within this document?	Current specification

Breadcrumbs sit above all four as a persistent statement of position, and are the primary means of moving back up a level: Home › Category › Sub-group › Specification.

5. User Flows

Four audiences use a specification portal, and their needs are close to orthogonal. The flows below are ordered by expected frequency.

5.1 Primary — known-item retrieval

The most common flow by a wide margin. A user knows what they want and is measuring the product on how few actions it takes.

```
Entry ─▶ ⌘K ─▶ type query ─▶ results (debounced, ranked) ─▶ ↑ ↓ ─▶ ↵
    └▶ Specification view ─▶ Download
```

- Target: three keystrokes to a result, no pointer required at any step.
- The palette opens over whatever view is current; it never navigates away before a selection is made.
- Escape returns the user exactly where they were, with focus restored to the trigger.

5.2 Secondary — exploratory browse

The user knows the category but not the specific record, or is comparing options.

```
Landing ─▶ Filter panel (category · status · tags)
  │       └▶ Result list updates in place, URL reflects state
  └▶ Sidebar tree ─▶ expand category ─▶ expand sub-group
        └▶ Specification view ◀─▶ prev / next within branch
```

- Filter state is held in the URL so that a filtered view is shareable and survives reload.
- Previous and next controls let a user move through a branch without returning to the list — the main weakness of the reference system's browse loop.
- The sidebar tree and the filter panel stay synchronised; selecting a category in one reflects in the other.

5.3 Tertiary — production handoff

A supplier or print partner who needs the artefact, not the page. This flow ends outside the application, and its success measure is whether the exported material is complete and unambiguous.

```
Specification view ─▶ Downloads bar
    ├─▶ Download file(s)
    ├─▶ Export PDF     (full record, print-optimised)
    ├─▶ Print view     (dedicated stylesheet)
    └─▶ Copy link      (deep link, confirmed by toast)
```

- The export must carry the full record — identity, status, version, technical detail, requirements and notes — not only what is visible on screen.
- Status and version appear on every exported page, so a printed sheet cannot be mistaken for current when it is not.

5.4 Administrative — authoring and maintenance

Lower frequency, higher consequence. Every step is reversible and every destructive action is soft.

```
Sign in ─▶ Admin mode ─▶ Management sheet
  ├─ Specs       list ─▶ new / edit ─▶ form ─▶ validate ─▶ save ─▶ toast
  │                  └▶ archive (soft) ─▶ restorable from Archive
  ├─ Categories  create · rename · reorder · nest sub-groups
  ├─ Users       invite · role · revoke
```

```
├─ Archive    browse ➔ restore | permanently delete (confirmed)
└─ Settings   integrations · storage · appearance
```

- Saving writes a revision record automatically — revision history is a by-product of authoring, never a separate chore, or it will not be maintained.
- Archiving is reversible. Permanent deletion exists only inside the Archive tab and requires explicit confirmation.
- Unsaved changes are guarded on navigation away from the form.

5.5 First-run — the empty catalogue

Worth designing deliberately rather than by default, because it is the state the platform ships in and the state you will spend your first sessions inside.

```
Empty portal ➔ "No specifications yet" ➔ Sign in ➔ Create first
      ↳➔ Category required ➔ inline category creation in the form
```

Every list, every section and every panel therefore needs a considered empty state with a clear next action, not a blank region. This is treated as a design-system concern in Phase 3, not as per-screen decoration.

6. Sitemap & Route Table

Hash routing is retained, matching the reference system and keeping the build deployable to GitHub Pages without server rewrite rules. The route abstraction is isolated so a switch to history routing later is a single-file change.

6.1 Sitemap

```
/
├── #/                          Landing — hero, search, filters, list
│
├── #/browse                    All specifications — filterable index
│   └── ?category= &tag= &status= &q= &sort= &view=
│
├── #/category/:categoryId      Category index
│   └── #/category/:categoryId/:groupId
│
├── #/spec/:specId              Specification detail  (split-pane)
│   ├── #/spec/:specId#technical      anchor
│   ├── #/spec/:specId#requirements   anchor
│   ├── #/spec/:specId#metadata       anchor
│   ├── #/spec/:specId#notes          anchor
│   ├── #/spec/:specId#attachments    anchor
│   ├── #/spec/:specId#related        anchor
│   ├── #/spec/:specId#history        anchor
│   └── #/spec/:specId/print          print view
│
├── #/search?q=                 Full search results
├── #/bookmarks                 Saved specifications
├── #/recent                    Recently viewed
│
├── #/admin                     Guarded — management sheet
│   ├── #/admin/specs           List · search · create · edit
│   │   ├── #/admin/specs/new
│   │   └── #/admin/specs/:specId/edit
│   ├── #/admin/categories      Categories and sub-groups
│   ├── #/admin/users           Users and roles
│   ├── #/admin/archive         Archived · restore · delete
│   └── #/admin/settings        Integrations · storage · appearance
│
└── #/404                       Not found
```

6.2 Route table

Route	View	Layout	Guard
#/	Landing	Shell	Public
#/browse	Specification index	Shell	Public
#/category/:id	Category index	Shell	Public
#/category/:id/:groupId	Sub-group index	Shell	Public
#/spec/:id	Specification detail	Shell + split pane	Public
#/spec/:id/print	Print view	Bare	Public
#/search	Search results	Shell	Public
#/bookmarks	Bookmarks	Shell	Public

Route	View	Layout	Guard
#/recent	Recently viewed	Shell	Public
#/admin/*	Management sheet	Shell + overlay	Authenticated
#/404	Not found	Shell	Public

The management sheet renders as an overlay above the current view rather than as a separate page, preserving the reference system's behaviour — but it is addressable, so an admin screen can be linked to and reloaded directly. That is a strict improvement over the current modal, which has no URL.

7. Confirmed Direction

Recorded from your Phase 1 responses. These constrain every subsequent phase.

Decision	Confirmed	Consequence
Admin scope	Full admin layer	Authentication, five-tab management sheet, specification editor, category manager, archive, users, settings and notifications are all in scope — wired to empty data so you can populate through the interface.
Detail layout	Split-pane, refined	The 50/50 data-and-image split is preserved as the product's signature. Refinement means typography, spacing, contrast and keyboard behaviour — not restructuring. The anchor rail is accommodated within the data panel rather than as a third column.
Data source	Local TypeScript files	Typed data in /data behind a service layer. Static build, deployable to GitHub Pages exactly as the current site is. An API or CMS can be substituted later by replacing one adapter.
Deliverable format	Word / PDF documents	Each phase is issued as a formal document for review and sign-off.

7.1 Note on the split-pane decision

One consequence is worth stating now rather than discovering in Phase 2. The reference specification view carries four content blocks. The platform you have described carries twelve sections, several of which — revision history, attachments, related specifications, metadata — are inherently long or tabular.

In a 50%-width panel, tabular content is tight. Three mitigations are proposed for your consideration in Phase 2, and they are compatible with each other:

4. The panel widens to 58% on viewports above 1440px, where the image pane has surplus space.
5. Revision history and attachments render as full-width sections beneath the split rather than inside the panel, so the split governs the upper fold only.
6. The image pane collapses to a rail on demand, temporarily giving the panel full width, with the state remembered.

The wireframes will show the split at its default state, with these variants presented alongside so you can choose against something concrete.

8. Phase 2 Scope & Approval

On approval of this document, Phase 2 produces low-fidelity wireframes — grayscale only, no branding, no colour, no decorative graphics, every field a placeholder.

8.1 Screens

Screen	Wireframe coverage
Landing	Header, logo placeholder, navigation, search bar, filter panel, specification list, footer
Browse / index	Filter panel, result list and grid variants, sort control, pagination, empty state
Category index	Breadcrumb, category header, sub-group sections, specification list
Specification detail	All twelve sections at Section 4.2, every one empty, with the three split-pane variants
Search results	Command palette overlay and full results page
Bookmarks / recent	List with empty state
Admin — five tabs	Specs list, category manager, users, archive, settings
Specification editor	Full form with every field, drop zones and validation positions
System states	Empty, loading, skeleton, error, 404

8.2 Breakpoints

Each screen wireframed at three widths, with adaptation annotated between them: desktop at 1440px, tablet at 768px, mobile at 390px.

8.3 Approval

Phase 2 will not begin until this document is approved, and no final UI will be produced until the Phase 2 wireframes are approved in turn. If any finding in Section 3 is contested — particularly the accessibility corrections, which change the smallest type sizes and the lightest text colours — raise it now, while it is cheap to change.

End of Phase 1. Deliverables 1–4 of 12 complete. Awaiting approval to proceed.